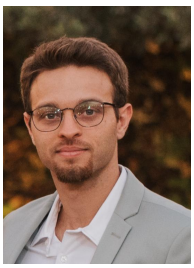
# Guy Bar-Shalom

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https://barsguy.github.io/



## Education

2022 – 2026 **PhD Computer Science.**
Technion - Israel Institute of Technology.
Advisors Prof. Ran El-Yaniv, and Prof. Haggai Maron.
*My research focuses on Geometric Deep Learning, leveraging data symmetries to enhance model learning. I explore applications in graph neural networks and uncertainty estimation in both vision models and large language models.*

2020 – 2022 **MSc Computer Science.**
Advisor Prof. Ran El-Yaniv.
Technion - Israel Institute of Technology.
*Uncertainty Estimation in Vision Models.*

2016 – 2020 **BSc Computer Science and Physics.**
Technion - Israel Institute of Technology.

## Employment

2025 **Meta.**
Research Intern
Worked on weight space models (paper under review at ICLR 2026)

2023 **Google-Verily.**
Research Intern
Worked on video analysis (published at WACV 2024)

2018 – present **Teaching assistant (Technion).**
*Algorithms 1.*
*Partial differential equations and Fourier series.*
*Introduction to Computer Science - C.*

## Selected Publications (* indicates equal contribution)

2026
- A Graph Meta-Network for Learning on Kolmogorov–Arnold Networks
  **Guy Bar-Shalom**, Ami Tavory, Itay Evron, Maya Bechler-Speicher, Ido Guy, Haggai Maron
  (ICLR 2026)
- Neural Message-Passing on Attention Graphs for Hallucination Detection
  Fabrizio Frasca*, **Guy Bar-Shalom***, Yftah Ziser, Haggai Maron
  (ICLR 2026)
- FS-KAN: Permutation Equivariant Kolmogorov-Arnold Networks via Function Sharing
  Ran Elbaz, **Guy Bar-Shalom**, Yam Eitan, Fabrizio Frasca, Haggai Maron
  (ICLR 2026)
- On The Expressive Power of GNN Derivatives
  Yam Eitan, Moshe Eliasof, Yoav Gelberg, Fabrizio Frasca, **Guy Bar-Shalom**, Haggai Maron
  (ICLR 2026)

## Selected Publications (* indicates equal contribution) (continued)

- Beyond Next Token Probabilities: Learnable, Fast Detection of Hallucinations and Data Contamination on LLM Output Distributions
  **Guy Bar-Shalom***, Fabrizio Frasca*, Derek Lim, Yoav Gelberg, Yftah Ziser, Ran El-Yaniv, Gal Chechik, Haggai Maron
  (AAAI 2026)
  A shorter version appeared @ Quantify Uncertainty and Hallucination in Foundation Models workshop @ ICLR 2025 (oral)

2025

- Beyond Token Probes: Hallucination Detection via Activation Tensors with ACT-ViT
  **Guy Bar-Shalom***, Fabrizio Frasca*, Yaniv Galron, Yftah Ziser, Haggai Maron
  (NeurIPS 2025)
- Balancing Efficiency and Expressiveness: Subgraph GNNs with Walk-Based Centrality
  Joshua Southern, Yam Eitan, **Guy Bar-Shalom**, Michael M. Bronstein, Haggai Maron, Fabrizio Frasca
  (ICML 2025)
- Topological Blindspots: Understanding and Extending Topological Deep Learning Through the Lens of Expressivity
  Yam Eitan, Yoav Gelberg, **Guy Bar-Shalom**, Fabrizio Frasca, Michael M. Bronstein, Haggai Maron
  Oral presentation
  (ICLR 2025)

2024

- A Flexible, Equivariant Framework for Subgraph GNNs via Graph Products and Graph Coarsening
  **Guy Bar-Shalom***, Yam Eitan*, Fabrizio Frasca, Haggai Maron
  (NeurIPS 2024)
  A shorter version appeared @ Symmetry and Geometry in Neural Representations workshop @ NeurIPS 2024 (oral) winning the best paper award
- Subgraphormer: Unifying Subgraph GNNs and Graph Transformers via Graph Products
  **Guy Bar-Shalom**, Beatrice Bevilacqua, Haggai Maron
  (ICML 2024)
- Weakly-supervised Representation Learning for Video Alignment and Analysis
  **Guy Bar-Shalom**, George Leifman, Michael Elad
  (WACV 2024)

2023

- Window-Based Distribution Shift Detection for Deep Neural Networks
  **Guy Bar-Shalom**, Yonatan Geifman, Ran El-Yaniv
  (NeurIPS 2023)

2022

- TransBoost: Improving the Best ImageNet Performance using Deep Transduction
  Omer Belhasin*, **Guy Bar-Shalom***, Ran El-Yaniv
  (NeurIPS 2022)

## Awards

2025

- Jacobs Qualcomm Excellence Scholarship
- Faculty Excellence Scholarship, Technion

2024

- best paper award @ Symmetry and Geometry in Neural Representations workshop @ NeurIPS 2024
- VATAT Prize for Academic Excellence, Technion

2018 – present

- Teaching Assistant Excellence award (4 times).

## Professional service

2024 — Co-organizer of iGDL: Israeli Geometric Deep Learning workshop

2022-present — Reviewer at NeurIPS, ICML, ICLR

## Teaching experience

2018 – present — Teaching Assistant at the Technion (department of Mathematics)
Course: *Partial Differential Equations and Fourier Series*
Average teaching score: 4.5/5.

2021 – present — Teaching Assistant at the Technion (department of Computer Science)
Course: *Algorithms 1*
Average teaching score: 4.6/5.

2021 — Teaching Assistant at the Technion (department of Computer Science)
Course: *Introduction to Computer Science*
Average teaching score: 4.4/5.

## Other

2012 — Played in the NCAA basketball league (USA), at Mercyhurst University (Erie, PA)

2012 – present — Semi-pro basketball player in Israel

2013-2016 — Israel Defense Forces (IDF), Staff Sergeant